

The magnitude response of low-pass filter in terms of gain and attenuation are shown in Figure 8.6.

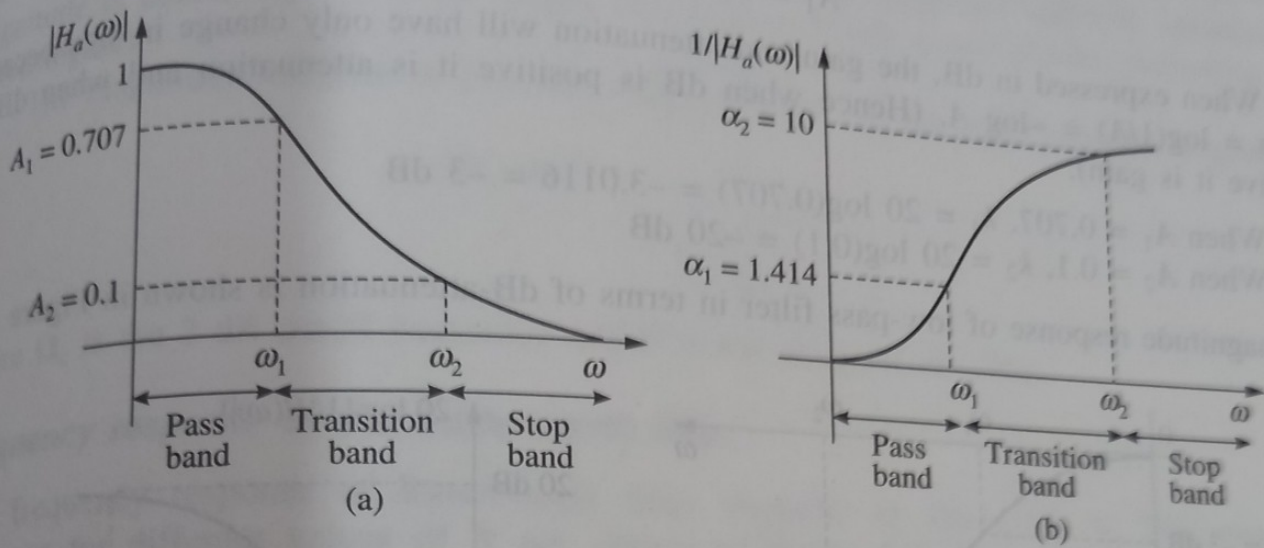


Figure 8.6 Magnitude response of low-pass filter (a) Gain vs ω and (b) Attenuation vs ω .

Let ω_1 = Passband frequency in rad/s.

ω_2 = Stopband frequency in rad/s.

Let the gain at the passband frequency ω_1 be A_1 and the gain at the stopband frequency ω_2 be A_2 , i.e.

$$A_1 = |H(\omega)|_{\omega=\omega_1} \quad \text{and} \quad A_2 = |H(\omega)|_{\omega=\omega_2}$$

The filter may be expressed in terms of the gain or attenuation at the edge frequencies. Let α_1 be the attenuation at the passband edge frequency ω_1 , and α_2 be the attenuation at the stopband edge frequency ω_2 .

$$\text{i.e.} \quad \alpha_1 = \frac{1}{A_1} = \frac{1}{|H(\omega)|_{\omega=\omega_1}} \quad \text{and} \quad \alpha_2 = \frac{1}{A_2} = \frac{1}{|H(\omega)|_{\omega=\omega_2}}$$

The maximum value of normalized gain is unity, so A_1 and A_2 are less than 1 and α_1 and α_2 are greater than 1. In Figure 8.6, A_1 is assumed as $1/\sqrt{2}$ and A_2 is assumed as 0.1. Hence $\alpha_1 = \sqrt{2} = 1.414$ and $\alpha_2 = 1/0.1 = 10$.

Another popular unit that is used for filter specification is dB. When the gain is expressed in dB, it will be a negative dB. When the attenuation is expressed in dB, it will be a positive dB.

Let k_1 = Gain in dB at a passband frequency ω_1

k_2 = Gain in dB at a stopband frequency ω_2

The gain can be converted into normal values as follows:

$$\begin{aligned} 20 \log A_1 &= k_1 \\ \log A_1 &= k_1/20 \\ A_1 &= 10^{k_1/20} \end{aligned}$$

$$\begin{aligned} 20 \log A_2 &= k_2 \\ \log A_2 &= k_2/20 \\ A_2 &= 10^{k_2/20} \end{aligned}$$

When expressed in dB, the gain and attenuation will have only change in sign because $\log \alpha = \log(1/A) = -\log A$. (Hence when dB is positive it is attenuation and when dB is negative it is gain).

$$\text{When } A_1 = 0.707, k_1 = 20 \log(0.707) = -3.0116 = -3 \text{ dB}$$

$$\text{When } A_2 = 0.1, k_2 = 20 \log(0.1) = -20 \text{ dB}$$

The magnitude response of low-pass filter in terms of dB-attenuation is shown in Figure 8.7.

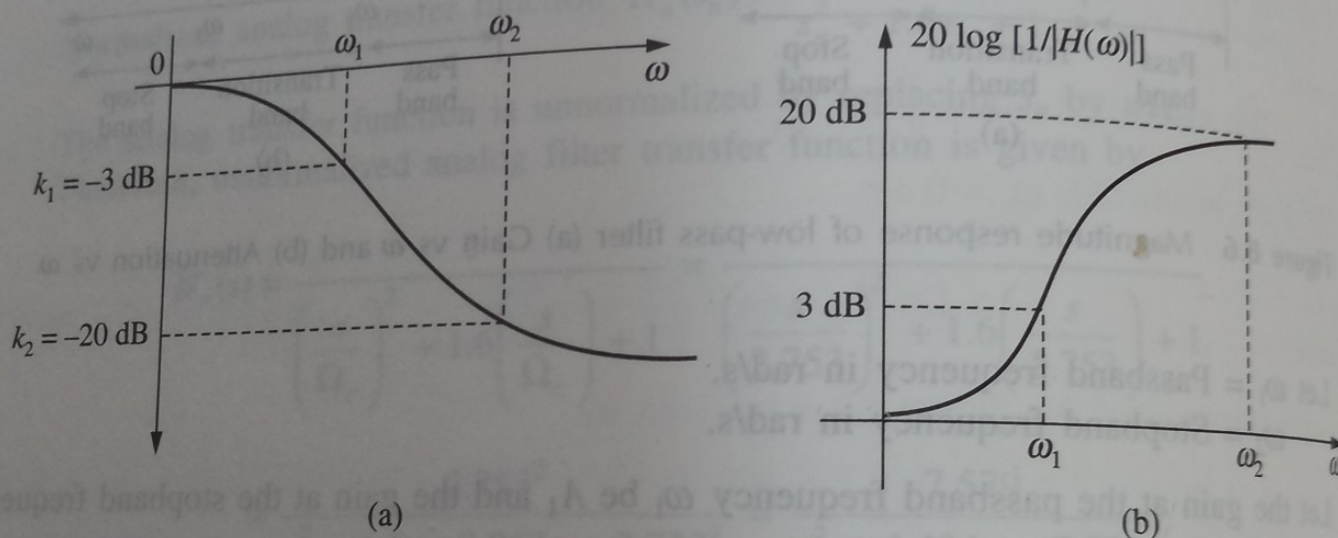


Figure 8.7 Magnitude response of low-pass filter (a) dB-Gain vs ω and (b) dB-attenuation vs ω .

Sometimes the specifications are given in terms of passband ripple δ_p and stopband ripple δ_s . In this case, the dB gain and attenuation can be estimated as follows:

$$k_1 = 20 \log (1 - \delta_p) \quad \alpha_1 = -20 \log (1 - \delta_p)$$

$$k_2 = 20 \log \delta_s \quad \alpha_2 = -20 \log \delta_s$$

If the ripples are specified in dB, then the minimum passband ripple is equal to k_1 and the negative of maximum passband attenuation is equal to k_2 .

8.7 DESIGN OF LOW-PASS DIGITAL BUTTERWORTH FILTER

The popular methods of designing IIR digital filter involves the design of equivalent analog filter and then converting the analog filter to digital filter. Hence to design a Butterworth IIR digital filter, first an analog Butterworth filter transfer function is determined using the given specifications. Then the analog filter transfer function is converted to a digital filter transfer function using either impulse invariant transformation or bilinear transformation.

Analog Butterworth filter

The analog Butterworth filter is designed by approximating the ideal frequency response using an error function. The error function is selected such that the magnitude is maximally flat in the passband and monotonically decreasing in the stopband. (Strictly speaking the magnitude is maximally flat at the origin, i.e., at $\Omega = 0$, and monotonically decreasing with increasing Ω).

The magnitude response of low-pass filter obtained by this approximation is given by

$$|H_a(\omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

where Ω_c is the 3 dB cutoff frequency and N is the order of the filter.

Frequency response of the Butterworth filter

The frequency response of Butterworth filter depends on the order N . The magnitude response for different values of N are shown in Figure 8.8. From Figure 8.8, it can be observed that the approximated magnitude response approaches the ideal response as the value of N increases. However, the phase response of the Butterworth filter becomes more nonlinear with increasing N .

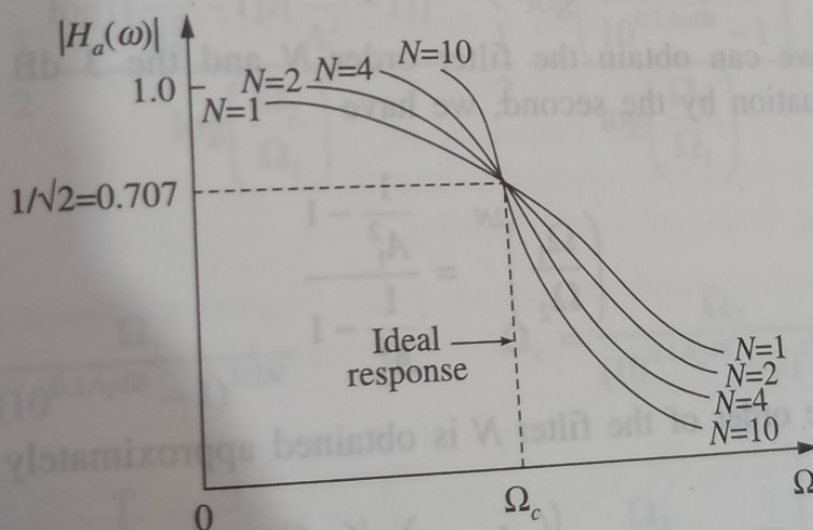


Figure 8.8 Magnitude response of Butterworth low-pass filter for various values of N .

Order of the filter

Since the frequency response of the filter depends on its order N , the order N has to be estimated to satisfy the given specifications.

Usually the specifications of the filter are given in terms of gain A or attenuation α at a passband or stopband frequency as given below:

$$\begin{aligned} A_1 \leq |H(\omega)| \leq 1, & \quad 0 \leq \omega \leq \omega_1 \\ |H(\omega)| \leq A_2, & \quad \omega_2 \leq \omega \leq \pi \end{aligned}$$

The order of the filter is determined as given below.

Let Ω_1 and Ω_2 be the analog filter edge frequencies corresponding to digital frequencies ω_1 and ω_2 . The values of Ω_1 and Ω_2 are obtained using the bilinear transformation or impulse invariant transformation.

$$A_1^2 \leq \frac{1}{1 + \left(\frac{\Omega_1}{\Omega_c}\right)^{2N}} \leq 1$$

and

$$\frac{1}{1 + \left(\frac{\Omega_2}{\Omega_c}\right)^{2N}} \leq A_2^2$$

These two equations can be written in the form

$$\left(\frac{\Omega_1}{\Omega_c}\right)^{2N} \leq \frac{1}{A_1^2} - 1$$

and

$$\left(\frac{\Omega_2}{\Omega_c}\right)^{2N} \geq \frac{1}{A_2^2} - 1$$

Assuming equality we can obtain the filter order N and the 3 dB cutoff frequency Ω_c . Dividing the first equation by the second, we have

$$\left(\frac{\Omega_1}{\Omega_2}\right)^{2N} = \frac{\frac{1}{A_1^2} - 1}{\frac{1}{A_2^2} - 1}$$

From this equation, the order of the filter N is obtained approximately as:

$$N = \frac{1}{2} \frac{\log \left\{ \left(\frac{1}{A_2^2} - 1 \right) / \left(\frac{1}{A_1^2} - 1 \right) \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

If N is not an integer, the value of N is chosen to be the next nearest integer. Also we can get

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$$

when parameters A_1 and A_2 are given in dB.

A_1 in dB is given by

$$A_1 \text{ dB} = -20 \log A_1$$

i.e.

$$\log A_1 = -\frac{A_1 \text{ dB}}{20}$$

or

$$A_1 = 10^{-\frac{A_1 \text{ dB}}{20}}$$

$\therefore$

$$\frac{1}{A_1^2} - 1 = \frac{1}{\left(10^{-\frac{A_1 \text{ dB}}{20}}\right)^2} - 1$$

i.e.

$$\frac{1}{A_1^2} - 1 = 10^{0.1 A_1 \text{ dB}} - 1$$

Similarly

$$\frac{1}{A_2^2} - 1 = 10^{0.1 A_2 \text{ dB}} - 1$$

$\therefore$

$$N = \frac{\frac{1}{2} \frac{\log\left\{\left[\frac{1}{A_2^2} - 1\right] / \left[\frac{1}{A_1^2} - 1\right]\right\}}{\log\left(\frac{\Omega_2}{\Omega_1}\right)}}{\frac{1}{2} \frac{\log\left(\frac{10^{0.1 A_2 \text{ dB}} - 1}{10^{0.1 A_1 \text{ dB}} - 1}\right)}{\log\left(\frac{\Omega_2}{\Omega_1}\right)}}$$

and Ω_c is given by

$$\Omega_c = \frac{\Omega_1}{(10^{0.1 A_1 \text{ dB}} - 1)^{1/2N}} \quad \text{or} \quad \Omega_c = \frac{\Omega_2}{(10^{0.1 A_2 \text{ dB}} - 1)^{1/2N}}$$

In fact,

$$\Omega_c = \frac{1}{2} \left[\frac{\Omega_1}{[10^{0.1 A_1 \text{ dB}} - 1]^{1/2N}} + \frac{\Omega_2}{[10^{0.1 A_2 \text{ dB}} - 1]^{1/2N}} \right]$$

Butterworth low-pass filter transfer function

The unnormalized transfer function of the Butterworth filter is usually written in factored form as:

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2} \quad (\text{when } N \text{ is even})$$

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \prod_{k=1}^{\frac{N-1}{2}} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2} \quad (\text{when } N \text{ is odd})$$

or

$$\text{where } b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

If s/Ω_c (where Ω_c is the 3 dB cutoff frequency of the low-pass filter) is replaced by s_n , then the normalized Butterworth filter transfer function is given by

$$H_a(s) = \prod_{k=1}^{N/2} \frac{1}{s_n^2 + b_k s_n + 1} \quad (\text{when } N \text{ is even})$$

or

$$H_a(s) = \frac{1}{s_n + 1} \prod_{k=1}^{\frac{N-1}{2}} \frac{1}{s_n^2 + b_k s_n + 1} \quad (\text{when } N \text{ is odd})$$

$$\text{where } b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

Design procedure for low-pass digital Butterworth IIR filter

The low-pass digital Butterworth filter is designed as per the following steps:

Let A_1 = Gain at a passband frequency ω_1

A_2 = Gain at a stopband frequency ω_2

Ω_1 = Analog frequency corresponding to ω_1

Ω_2 = Analog frequency corresponding to ω_2

Step 1 Choose the type of transformation, i.e., either bilinear or impulse invariant transformation.

Step 2 Calculate the ratio of analog edge frequencies Ω_2/Ω_1 .

For bilinear transformation

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2}, \quad \Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} \quad \therefore \frac{\Omega_2}{\Omega_1} = \frac{\tan \omega_2/2}{\tan \omega_1/2}$$

For impulse invariant transformation,

$$\Omega_1 = \frac{\omega_1}{T}, \quad \Omega_2 = \frac{\omega_2}{T} \quad \therefore \frac{\Omega_2}{\Omega_1} = \frac{\omega_2}{\omega_1}$$

Step 3 Decide the order N of the filter. The order N should be such that

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

Choose N such that it is an integer just greater than or equal to the value obtained above.

If ϕ is the angle of a valid pole w.r.t. the X-axis, then the pole and its conjugate are located at $[\omega_c(\cos \phi \pm j \sin \phi)]$.

Properties of Butterworth filters

1. The Butterworth filters are all pole designs (i.e. the zeros of the filters exist at ∞).
2. The filter order N completely specifies the filter.

Step 4 Calculate the analog cutoff frequency $\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$

For bilinear transformation $\Omega_c = \frac{\frac{2}{T} \tan \omega_1/2}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$

For impulse invariant transformation $\Omega_c = \frac{\omega_1/T}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}}$

Step 5 Determine the transfer function of the analog filter. Let $H_a(s)$ be the transfer function of the analog filter. When the order N is even, for unity dc gain filter, $H_a(s)$ is given by

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

When the order N is odd, for unity dc gain filter, $H_a(s)$ is given by

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \prod_{k=1}^{N-1} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

The coefficient b_k is given by

$$b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

For normalized case, $\Omega_c = 1$ rad/s

Step 6 Using the chosen transformation, transform the analog filter transfer function $H_a(s)$ to digital filter transfer function $H(z)$.

Step 7 Realize the digital filter transfer function $H(z)$ by a suitable structure.

3. The magnitude response approaches the ideal response as the value of N increases.
4. The magnitude is maximally flat at the origin.
5. The magnitude is monotonically decreasing function of Ω .
6. At the cutoff frequency Ω_c , the magnitude of normalized Butterworth filter is $1/\sqrt{2}$. Hence the dB magnitude at the cutoff frequency will be 3 dB less than the maximum value.

EXAMPLE 8.17 Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.9 \leq |H(\omega)| \leq 1; \quad 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(\omega)| \leq 0.2; \quad \frac{3\pi}{4} \leq \omega \leq \pi$$

with $T = 1$ s

Solution: The Butterworth digital filter is designed as per the following steps.
From the given specification, we have

$$A_1 = 0.9 \text{ and } \omega_1 = \frac{\pi}{2}$$

$$A_2 = 0.2 \text{ and } \omega_2 = \frac{3\pi}{4} \quad \text{and } T = 1 \text{ s}$$

Step 1 Choice of the type of transformation

Here the bilinear transformation is already specified.

Step 2 Determination of the ratio of the analog filter's edge frequencies, Ω_2/Ω_1

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = \frac{2}{1} \tan \left[\frac{(3\pi/4)}{2} \right] = 2 \tan \frac{3\pi}{8} = 4.828$$

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = \frac{2}{1} \tan \left[\frac{(\pi/2)}{2} \right] = 2 \tan \frac{\pi}{4} = 2$$

$$\therefore \frac{\Omega_2}{\Omega_1} = \frac{4.828}{2} = 2.414$$

Step 3 Determination of the order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] \left/ \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.2)^2} - 1 \right] / \left[\frac{1}{(0.9)^2} - 1 \right] \right\}}{\log 1.207}$$

$$\geq \frac{1}{2} \frac{\log \{24/0.2345\}}{\log 2.414} \geq 2.626$$

Since $N \geq 2.626$, choose $N = 3$.

Step 4 Determination of the analog cutoff frequency Ω_c (i.e., -3 dB frequency)

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{2}{\left[\frac{1}{0.9^2} - 1 \right]^{1/2 \times 3}} = 2.5467$$

Step 5 Determination of the transfer function of the analog Butterworth filter $H_a(s)$

For odd N , we have $H_a(s) = \frac{\Omega_c}{s + \Omega_c} \prod_{k=1}^{\frac{N-1}{2}} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$

where $b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$

For $N = 3$, we have

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2}$$

where $b_1 = 2 \sin \left[\frac{(2 \times 1 - 1)\pi}{2 \times 3} \right] = 2 \sin \frac{\pi}{6} = 1$

Therefore, $H_a(s) = \left(\frac{2.5467}{s + 2.5467} \right) \left(\frac{(2.5467)^2}{s^2 + 1(2.5467)s + (2.5467)^2} \right)$

Step 6 Conversion of $H_a(s)$ into $H(z)$

Since bilinear transformation is to be used, the digital filter transfer function is:

$$H(z) = H_a(s) \Big|_{s=2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s=2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \left(\frac{2.5467}{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2.5467} \right) \left(\frac{(2.5467)^2}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 2.5467 \left[2 \frac{1-z^{-1}}{1+z^{-1}} \right] + (2.5467)^2} \right)$$

$$= \frac{0.2332(1+z^{-1})^3}{1+0.4394z^{-1}+0.3845z^{-2}+0.0416z^{-3}}$$

EXAMPLE 8.18 Design a digital Butterworth filter satisfying the following constraints:

$$0.8 \leq |H(\omega)| \leq 1; \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(\omega)| \leq 0.2; \quad 0.32\pi \leq \omega \leq \pi$$

with $T = 1$ s. Apply impulse invariant transformation.

Solution: From the given specifications, we have

$$A_1 = 0.8 \quad \omega_1 = 0.2\pi$$

$$A_2 = 0.2 \quad \omega_2 = 0.32\pi \quad \text{and } T = 1 \text{ s}$$

The Butterworth IIR digital filter is designed as per the following steps.

Step 1 Choice of the type of transformation

Here, the impulse invariant transformation is already specified.

Step 2 Determination of the ratio of analog filter's edge frequencies, Ω_2/Ω_1

$$\Omega_2 = \frac{\omega_2}{T} = \frac{0.32\pi}{1} = 0.32\pi$$

$$\Omega_1 = \frac{\omega_1}{T} = \frac{0.2\pi}{1} = 0.2\pi$$

$$\frac{\Omega_2}{\Omega_1} = \frac{0.32\pi}{0.2\pi} = 1.6$$

Step 3 Determination of the order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] \left/ \left[\frac{1}{A_1^2} - 1 \right] \right. \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{0.2^2} - 1 \right] \left/ \left[\frac{1}{0.8^2} - 1 \right] \right. \right\}}{\log 1.6}$$

$$\geq \frac{1}{2} \frac{\log \{24/0.5625\}}{\log 1.6} \geq 3.9931$$

So the order of the filter $N \geq 3.9931$. Choose $N = 4$.

Step 4 Determination of the analog cutoff frequency Ω_c (i.e., -3 dB frequency)

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1\right]^{1/2N}} = \frac{0.2\pi}{\left[\frac{1}{0.8^2} - 1\right]^{1/2 \times 4}} = 0.675 \text{ rad/s}$$

Step 5 Determination of the transfer function of analog low-pass Butterworth filter. The transfer function of the low-pass filter for even values of N is:

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

$$b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

where

Here $N = 4$; $\therefore k = 1, 2$

$$\text{When } k = 1, b_k = b_1 = 2 \sin \left[\frac{(2-1)\pi}{2 \times 4} \right] = 0.765$$

$$\text{When } k = 2, b_k = b_2 = 2 \sin \left[\frac{(2 \times 2 - 1)\pi}{2 \times 4} \right] = 1.848$$

$$\therefore H_a(s) = \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2} \times \frac{\Omega_c^2}{s^2 + b_2 \Omega_c s + \Omega_c^2}$$

$$= \frac{(0.675)^2}{s^2 + (0.765 \times 0.675)s + 0.675^2} \times \frac{(0.675)^2}{s^2 + (1.848 \times 0.675)s + (0.675)^2}$$

$$= \frac{0.2076}{(s^2 + 0.516s + 0.456)(s^2 + 1.247s + 0.456)}$$

Step 6 Determination of the digital filter transfer function $H(z)$

By partial fraction expansion, $H_a(s)$ can be expressed as:

$$H_a(s) = \frac{0.2076}{(s^2 + 0.516s + 0.456)(s^2 + 1.247s + 0.456)}$$

$$= \frac{As + B}{s^2 + 0.516s + 0.456} + \frac{Cs + D}{s^2 + 1.247s + 0.456}$$

On cross multiplying the above equation and simplifying, we get

$$0.2076 = (A + C)s^3 + (1.247A + B + 0.516C + D)s^2 + (0.456A + 1.247B + 0.456C + 0.516D)s + (0.456B + 0.456D)$$

On solving, we get

$$A = -0.622, B = -0.321, C = 0.622 \text{ and } D = 0.776$$

Therefore, $H_a(s)$ can be written as:

$$\begin{aligned}
 H_a(s) &= \frac{-0.622s - 0.321}{s^2 + 0.516s + 0.456} + \frac{0.622s + 0.776}{s^2 + 1.247s + 0.456} \\
 &= \frac{-0.622(s + 0.516)}{(s^2 + 2 \times 0.258s + 0.258^2) + (\sqrt{0.456 - 0.258^2})^2} \\
 &\quad + \frac{0.622(s + 1.248)}{(s^2 + 2 \times 0.624s + 0.624^2) + (\sqrt{0.456 - 0.624^2})^2} \\
 &= \frac{-0.622(s + 0.258 + 0.258)}{(s + 0.258)^2 + (0.624)^2} + \frac{0.622(s + 0.624 + 0.624)}{(s + 0.624)^2 + (0.258)^2} \\
 &= -0.622 \frac{s + 0.258}{(s + 0.258)^2 + (0.624)^2} - 0.257 \frac{0.624}{(s + 0.258)^2 + (0.624)^2} \\
 &\quad + 0.622 \frac{s + 0.624}{(s + 0.624)^2 + (0.258)^2} + 1.504 \frac{0.258}{(s + 0.624)^2 + (0.258)^2}
 \end{aligned}$$

The analog transfer function of the above equation can be transformed to digital transfer function using the following standard impulse invariant transformations.

$$\begin{aligned}
 \frac{s + a}{(s + a)^2 + b^2} &\rightarrow \frac{1 - e^{-aT}(\cos bT)z^{-1}}{1 - 2e^{-aT}(\cos bT)z^{-1} + e^{-2aT}z^{-2}} \\
 \frac{b}{(s + a)^2 + b^2} &\rightarrow \frac{e^{-aT}(\sin bT)z^{-1}}{1 - 2e^{-aT}(\cos bT)z^{-1} + e^{-2aT}z^{-2}}
 \end{aligned}$$

Taking $T = 1$ s, the above transformation can be applied to $H_a(s)$ to get $H(z)$.

$$\begin{aligned}
 H(z) &= -0.622 \frac{1 - e^{-0.258}(\cos 0.624)z^{-1}}{1 - 2e^{-0.258}(\cos 0.624)z^{-1} + e^{-2 \times 0.258}z^{-2}} \\
 &\quad - 0.257 \frac{e^{-0.258}(\sin 0.624)z^{-1}}{1 - 2e^{-0.258}(\cos 0.624)z^{-1} + e^{-2 \times 0.258}z^{-2}} \\
 &\quad + 0.622 \frac{1 - e^{-0.624}(\cos 0.258)z^{-1}}{1 - 2e^{-0.624}(\cos 0.258)z^{-1} + e^{-2 \times 0.624}z^{-2}} \\
 &\quad + 1.504 \frac{e^{-0.624}(\sin 0.258)z^{-1}}{1 - 2e^{-0.624}(\cos 0.258)z^{-1} + e^{-2 \times 0.624}z^{-2}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{-0.622 + 0.39z^{-1}}{1 - 1.254z^{-1} + 0.597z^{-2}} + \frac{-0.116z^{-1}}{1 - 1.254z^{-1} + 0.597z^{-2}} \\
 &+ \frac{0.622 - 0.322z^{-1}}{1 - 1.036z^{-1} + 0.287z^{-2}} + \frac{0.206z^{-1}}{1 - 1.036z^{-1} + 0.287z^{-2}} \\
 &= \frac{-0.622 + 0.274z^{-1}}{1 - 1.254z^{-1} + 0.597z^{-2}} + \frac{0.622 - 0.116z^{-1}}{1 - 1.036z^{-1} + 0.287z^{-2}} \\
 &= \frac{0.0224z^{-1} + 0.0544z^{-2} + 0.0094z^{-3}}{1 - 2.29z^{-1} + 2.1831z^{-2} - 0.977z^{-3} + 0.1713z^{-4}}
 \end{aligned}$$

EXAMPLE 8.19 Design a low-pass Butterworth digital filter to give response of 3 dB or less for frequencies upto 2 kHz and an attenuation of 20 dB or more beyond 4 kHz. Use the bilinear transformation technique and obtain $H(z)$ of the desired filter.

Solution: The specifications of the desired filter are given in terms of dB attenuation and frequency in Hz. First the gain is to be expressed as a numerical value and frequency in rad/s.

Here attenuation at passband frequency (ω_1) = 3 dB

Therefore, gain at passband edge frequency (ω_1) is $k_1 = -3$ dB

$$A_1 = 10^{k_1/20} = 10^{-3/20} = 0.707 = \frac{1}{\sqrt{2}}$$

Attenuation at stopband frequency (ω_2) = 20 dB

Therefore, gain at stopband edge frequency (ω_2) is $k_2 = -20$ dB

$$A_2 = 10^{k_2/20} = 10^{-20/20} = 0.1$$

Passband edge frequency = 2 kHz,

Stopband edge frequency = 4 kHz,

The design is performed as given below.

Let the sampling frequency be 10000 Hz.

$$\text{Normalized } \omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \frac{2000}{10000} = 0.4\pi$$

$$\text{Normalized } \omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \frac{4000}{10000} = 0.8\pi$$

Step 1 Bilinear transformation is chosen

Step 2 Ratio of analog filter edge frequencies Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = \frac{2}{T} \tan \frac{0.4\pi}{2} = 14530.8 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = \frac{2}{T} \tan \frac{0.8\pi}{2} = 61553.6 \text{ rad/s}$$

$$\therefore \frac{\Omega_2}{\Omega_1} = \frac{\tan \frac{\omega_2}{2}}{\tan \frac{\omega_1}{2}} = \frac{\tan 0.4\pi}{\tan 0.2\pi} = 4.236$$

Step 3 Order of the filter

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \frac{\Omega_2}{\Omega_1}}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.1)^2} - 1 \right] / \left[\frac{1}{(1/\sqrt{2})^2} - 1 \right] \right\}}{\log 4.236}$$

$$\geq \frac{1}{2} \frac{\log [99/1]}{\log 4.236} \geq 1.59$$

$$\therefore N = 2$$

Step 4 Analog cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{1.4530}{\left[\frac{1}{(1/\sqrt{2})^2} - 1 \right]^{1/2 \times 2}} = 1.4530$$

$$\text{Unnormalized } \Omega_c = f_s \times 1.4530 = 14530 \text{ rad/s}$$

Step 5 Transfer function $H_a(s)$

$$\text{For } N = 2, H_a(s) = \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2}$$

$$\text{where } b_1 = 2 \sin \left[\frac{(2 \times 1 - 1)\pi}{2 \times 2} \right] = 2 \sin \frac{\pi}{4} = 1.414$$

$$\therefore H_a(s) = \frac{(14530)^2}{s^2 + 1.414 \times 14530 s + (14530)^2}$$

$$= \frac{2.1112 \times 10^8}{s^2 + 20545.42 s + 2.1112 \times 10^8}$$

Step 6 Conversion of $H_a(s)$ into $H(z)$

$$H(z) = H_a(s) \Big|_{s=\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s=20000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \frac{2.112 \times 10^8}{\left[20 \times 10^3 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 2.0545 \times 10^4 \times 20 \times 10^3 \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 2.112 \times 10^8}$$

$$= \frac{0.528}{2.5552 - 0.946z^{-1} + 0.5008z^{-2}}$$

EXAMPLE 8.20 Design a low-pass Butterworth filter using the bilinear transformation method for satisfying the following constraints:

Passband: 0–400 Hz

Passband ripple: 2 dB

Sampling frequency: 10 kHz

Stopband: 2.1–4 kHz

Stopband attenuation: 20 dB

Solution: Given

$$\alpha_1 = 2 \text{ dB}, \quad \therefore k_1 = -2 \text{ dB} \quad \text{and} \quad A_1 = 10^{k_1/20} = 10^{-2/20} = 0.794$$

$$\alpha_2 = 20 \text{ dB}, \quad \therefore k_2 = -20 \text{ dB} \quad \text{and} \quad A_2 = 10^{k_2/20} = 10^{-20/20} = 0.1$$

Step 1 Type of transformation

Bilinear transformation is already specified.

Step 2 Ratio of analog edge frequencies Ω_2/Ω_1 .

Here $f_s = 10 \text{ kHz}$

Passband edge frequency $f_1 = 400 \text{ Hz}$

Stopband edge frequency $f_2 = 2.1 \text{ kHz}$

Normalizing the frequencies, we have

$$\omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \times \frac{400}{10000} = 0.25 \text{ rad}$$

$$\omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \times \frac{2100}{10000} = 1.319 \text{ rad}$$

Therefore, the analog filter edge frequencies are:

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 10000 \tan \frac{0.25}{2} = 2513.102 \text{ rad/s}$$

and $\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 10000 \tan \frac{1.319}{2} = 15,506.08 \text{ rad/s}$

$$\frac{\Omega_2}{\Omega_1} = \frac{15506.08}{2513.102} = 6.1703$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left\{ \frac{10^{0.1A_2 \text{ dB}} - 1}{10^{0.1A_1 \text{ dB}} - 1} \right\}}{\log (6.1703)}$$

$$\text{i.e. } N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.1)^2} - 1 \right] / \left[\frac{1}{(0.794)^2} - 1 \right] \right\}}{\log (6.1703)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left\{ \frac{10^{0.1 \times 20 \text{ dB}} - 1}{10^{0.1 \times 2 \text{ dB}} - 1} \right\}}{\log (6.1703)}$$

$$\text{i.e. } N \geq 1.409 \approx 2 \quad \text{or} \quad N \geq 1.410 \approx 2$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_2}{\left[\frac{1}{A_2^2} - 1 \right]^{1/2N}} \quad \text{or} \quad \Omega_c = \frac{\Omega_2}{[10^{0.1A_2 \text{ dB}} - 1]^{1/2N}}$$

i.e.

$$\Omega_c = \frac{15506.08}{\left[\frac{1}{(0.1)^2} - 1 \right]^{1/2 \times 2}} = 4915.7 \quad \text{or} \quad \Omega_c = \frac{15506.08}{[10^{0.1 \times 20} - 1]^{1/2 \times 2}} = 4915.788 \text{ rad/s}$$

Step 5 The system function $H_a(s)$

$$\begin{aligned} H_a(s) &= \frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2} \quad \text{where } b_1 = 2 \sin \frac{(2 \times 1 - 1)\pi}{2 \times 2} = 1.414 \\ &= \frac{(4915.788)^2}{s^2 + 1.414 \times 4915.788 s + (4915.788)^2} \\ &= \frac{2.416 \times 10^7}{s^2 + 6950.92 s + 2.416 \times 10^7} \end{aligned}$$

Step 6 Digital transfer function $H(z)$

$$H(z) = H_a(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \frac{2.416 \times 10^7}{\left[\frac{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 6950.92 \times \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 2.416 \times 10^7}$$

$$= \frac{2.416 \times 10^7}{\left[20000 \times \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 6950.92 \times 20000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2.416 \times 10^7}$$

$$= \frac{0.042 + 0.085z^{-1} + 0.042z^{-2}}{1 - 1.335z^{-1} + 0.506z^{-2}}$$

The poles are given by $P_k = \pm (\Omega_c) \left[e^{j \frac{(2k+N+1)\pi}{2N}} \right]$, $k = 0, 1 < N$

$$\therefore P_0 = \pm (\Omega_c) e^{j \left(\frac{3\pi}{4} \right)} = 4.915.788(-0.707 + j0.707) = -3475.6 + j3475.46$$

$$P_1 = \pm (\Omega_c) e^{j \left(\frac{5\pi}{4} \right)} = -3475.6 - j3475.6$$

EXAMPLE 8.21 A digital low-pass filter is required to meet the following specifications.

Passband attenuation ≤ 1 dB

Passband edge = 4 kHz

Stopband attenuation ≥ 40 dB

Stopband edge = 8 kHz

Sampling rate = 24 kHz

The filter is to be designed by performing the bilinear transformation on an analog system function. Design the Butterworth filter.

Solution: Given $\alpha_1 = 1$ dB, $\therefore k_1 = -1$ dB and $A_1 = 10^{k_1/20} = 10^{-1/20} = 0.8912$

$\alpha_2 = 40$ dB, $\therefore k_2 = -40$ dB and $A_2 = 10^{k_2/20} = 10^{-40/20} = 0.01$

Since $f_s = 24$ kHz, normalized angular frequencies are:

$$f_1 = 4 \text{ kHz}, \therefore \omega_1 = \frac{2\pi f_1}{f_s} = 2\pi \times \frac{4000}{24000} = 1.047 \text{ rad/s}$$

$$f_2 = 8 \text{ kHz}, \therefore \omega_2 = \frac{2\pi f_2}{f_s} = 2\pi \times \frac{8000}{24000} = 2.094 \text{ rad/s}$$

The Butterworth filter is designed as follows:

Step 1 Type of transformation

Bilinear transformation is already specified.

Step 2 Ratio of analog edge frequencies, Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 24000 \tan \frac{1.047}{2} = 27706.49 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 24000 \tan \frac{2.094}{2} = 83100.52 \text{ rad/s}$$

$$\therefore \frac{\Omega_2}{\Omega_1} = \frac{83000.52}{27706.49} = 2.9957$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2 / \Omega_1)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left(\frac{10^{0.1 A_2 \text{ dB}} - 1}{10^{0.1 A_1 \text{ dB}} - 1} \right)}{\log (\Omega_2 / \Omega_1)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.01)^2} - 1 \right] / \left[\frac{1}{(0.8912)^2} - 1 \right] \right\}}{\log (2.9957)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log \left(\frac{10^{0.1 \times 40} - 1}{10^{0.1 \times 1} - 1} \right)}{\log (2.9957)}$$

$$\geq \frac{1}{2} \frac{\log \{9999/0.2590\}}{\log (2.9957)} \quad \text{or} \quad N \geq \frac{1}{2} \frac{\log (9999/0.2589)}{\log (2.9957)}$$

$$\geq \frac{1}{2} \frac{4.586}{0.476} \quad \text{or} \quad N \geq \frac{1}{2} \frac{4.586}{0.476}$$

$$\geq 4.8 \approx 5$$

$$\text{or} \quad N \geq 4.8 \approx 5$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} \quad \text{or} \quad \Omega_c = \frac{\Omega_1}{[10^{0.1 A_1 \text{ dB}} - 1]^{1/2N}}$$

$$= \frac{27706.49}{\left[\frac{1}{(0.8912)^2} - 1 \right]^{1/2 \times 5}} \quad \text{or} \quad \Omega_c = \frac{27706.49}{[10^{0.1 \times 1} - 1]^{1/2 \times 5}}$$

$$= 31,715 \text{ rad/s}$$

$$\text{or} \quad \Omega_c = 31,715 \text{ rad/s}$$

Step 5 Analog filter transfer function $H_a(s)$

$$\text{For } N = 5, H_a(s) = \left(\frac{\Omega_c}{s + \Omega_c} \right) \left(\frac{\Omega_c^2}{s^2 + b_1 \Omega_c s + \Omega_c^2} \right) \left(\frac{\Omega_c^2}{s^2 + b_2 \Omega_c s + \Omega_c^2} \right)$$

where

$$b_1 = 2 \sin \left(\frac{(2 \times 1 - 1) \pi}{2N} \right) = 2 \sin \frac{\pi}{10} = 0.618$$

$$b_2 = 2 \sin \left(\frac{(2 \times 2 - 1) \pi}{10} \right) = 2 \sin \frac{3\pi}{10} = 1.618$$

$$\therefore H_a(s) = \left(\frac{31708}{s + 31708} \right) \left(\frac{(31708)^2}{s^2 + 0.618 \times 31708 s + (31708)^2} \right) \\ \left(\frac{(31708)^2}{s^2 + 1.618 \times 31708 s + (31708)^2} \right)$$

Step 6 Digital filter function $H(z)$
Using the bilinear transformation, we have

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \Big|_{s = 48000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \left[\left(\frac{31708}{s + 31708} \right) \left(\frac{(31708)^2}{s^2 + 0.618 \times 31708 s + (31708)^2} \right) \right. \\ \left. \left(\frac{(31708)^2}{s^2 + 1.618 \times 31708 s + (31708)^2} \right) \right] \Big|_{s = 48000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

EXAMPLE 8.22 Design a digital IIR low-pass filter with passband edge at 1000 Hz and stopband edge at 1500 Hz for a sampling frequency of 5000 Hz. The filter is to have passband ripple of 0.5 dB and a stopband ripple below 30 dB. Design a Butterworth filter using the bilinear transformation.

Solution: Given $f_s = 5000$ Hz, the normalized frequencies are given as:

$$\text{Passband edge } f_1 = 1000 \text{ Hz, } \therefore \omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \times \frac{1000}{5000} = 0.4\pi \text{ rad/s}$$

$$\text{Stopband edge } f_2 = 1500 \text{ Hz, } \therefore \omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \times \frac{1500}{5000} = 0.6\pi \text{ rad/s}$$

$$\alpha_1 = 0.5 \text{ dB, } \therefore k_1 = -0.5 \text{ dB and } A_1 = 10^{k_1/20} = 10^{-0.5/20} = 0.9446$$

$$\alpha_2 = 30 \text{ dB, } \therefore k_2 = -30 \text{ dB and } A_2 = 10^{k_2/20} = 10^{-30/20} = 0.0316$$

The Butterworth filter is designed as follows:

Step 1 Type of transformation.
Bilinear transformation is to be used.

Step 2 Ratio of analog filter edge frequencies, Ω_2/Ω_1

$$\Omega_1 = \frac{2}{T} \tan \frac{\omega_1}{2} = 2 \times 5000 \tan \frac{0.4\pi}{2} = 7265.425 \text{ rad/s}$$

$$\Omega_2 = \frac{2}{T} \tan \frac{\omega_2}{2} = 2 \times 5000 \tan \frac{0.6\pi}{2} = 13763.819 \text{ rad/s}$$

$$\frac{\Omega_2}{\Omega_1} = \frac{13763.819}{7265.425} = 1.8944$$

Step 3 Order of the filter N

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] \middle/ \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2/\Omega_1)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.0316)^2} - 1 \right] \middle/ \left[\frac{1}{(0.9446)^2} - 1 \right] \right\}}{\log (1.844)}$$

$$\geq \frac{1}{2} \frac{\log \{1000.44/0.1207\}}{\log (1.844)}$$

$$\geq 7.35 \approx 8$$

Step 4 The cutoff frequency Ω_c

$$\Omega_c = \frac{\Omega_1}{\left[\frac{1}{A_1^2} - 1 \right]^{1/2N}} = \frac{7265.425}{\left[\frac{1}{0.9446^2} - 1 \right]^{1/2 \times 8}} = 8292 \text{ rad/s}$$

Step 5 The system function $H_a(s)$

$$H_a(s) = \prod_{k=1}^{N/2} \frac{\Omega_c^2}{s^2 + b_k \Omega_c s + \Omega_c^2}$$

$$\text{where } b_k = 2 \sin \left[\frac{(2k-1)\pi}{2N} \right]$$

$$b_1 = 2 \sin \left[\frac{\pi}{16} \right] = 0.390$$

$$b_2 = 2 \sin \left[\frac{3\pi}{16} \right] = 1.111$$

$$b_3 = 2 \sin \left[\frac{5\pi}{16} \right] = 1.662$$

$$b_4 = 2 \sin \left[\frac{7\pi}{16} \right] = 1.961$$

$$H_a(s) = \left(\frac{(8292)^2}{s^2 + 0.39 \times 8292s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 1.111 \times 8292s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 1.662 \times 8292s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 1.961 \times 8292s + (8292)^2} \right)$$

Step 6 Digital filter function $H(z)$
Using the bilinear transformation, we have

$$H(z) = H_a(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = H_a(s) \bigg|_{s = 10000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$H(z) = \left[\left(\frac{(8292)^2}{s^2 + 3233.8s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 9212.4s + (8292)^2} \right) \right] \left[\left(\frac{(8292)^2}{s^2 + 13781.3s + (8292)^2} \right) \left(\frac{(8292)^2}{s^2 + 16260.6s + (8292)^2} \right) \right] \bigg|_{s = 10000 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

EXAMPLE 8.23 Find the filter order for the following specifications:

$$\sqrt{0.5} \leq |H(\omega)| \leq 1 \quad 0 \leq \omega \leq \pi/2$$

$$|H(\omega)| \leq 0.2 \quad 3\pi/4 \leq \omega \leq \pi$$

with $T = 1$ s. Use the impulse invariant method.

Solution: Given $A_1 = \sqrt{0.5}$, $\omega_1 = \frac{\pi}{2}$

$$A_2 = 0.2, \quad \omega_2 = \frac{3\pi}{4}$$

$$T = 1 \text{ s.}$$

The impulse invariant transformation is to be used.

$$\frac{\Omega_2}{\Omega_1} = \frac{\omega_2/T}{\omega_1/T} = \frac{\omega_2}{\omega_1} = \frac{3\pi/4}{\pi/2} = 1.5$$

Order of the low-pass Butterworth filter N

$$\begin{aligned} N &\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log (\Omega_2 / \Omega_1)} \\ &\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.2)^2} - 1 \right] / \left[\frac{1}{(\sqrt{0.5})^2} - 1 \right] \right\}}{\log (1.5)} \\ &\geq \frac{1}{2} \frac{\log (24)}{\log (1.5)} \geq 3.919 \approx 4 \end{aligned}$$

EXAMPLE 8.24 Determine the order and the poles of a low-pass Butterworth filter that has a -3dB bandwidth of 500 Hz and an attenuation of 40 dB at 1000 Hz.

Solution: Given

Passband edge frequency $f_1 = 500$ Hz, $\therefore \omega_1 = 2\pi f_1 = 1000\pi$

Gain at passband edge $k_1 = -3$ dB, $\therefore A_1 = 10^{k_1/20} = 10^{-3/20} = 0.707$

Stopband edge frequency $f_2 = 1000$ Hz, $\therefore \omega_2 = 2\pi f_2 = 2000\pi$

Gain at stopband edge $k_2 = -40$ dB, $\therefore A_2 = 10^{k_2/20} = 10^{-40/20} = 0.01$

Let the sampling frequency $f_s = 2000$ Hz.

The normalized frequencies are:

$$\omega_1 = 2\pi \frac{f_1}{f_s} = 2\pi \frac{500}{2000} = 0.5\pi$$

$$\omega_2 = 2\pi \frac{f_2}{f_s} = 2\pi \frac{1000}{2000} = \pi$$

For impulse invariant transformation,

$$\frac{\Omega_2}{\Omega_1} = \frac{\omega_2}{\omega_1} = 2$$

Therefore, order of the filter is:

$$N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{(0.01)^2} - 1 \right] / \left[\frac{1}{(0.707)^2} - 1 \right] \right\}}{\log 2}$$

$$\geq \frac{1}{2} \frac{\log \{999/1\}}{\log(2)} \geq 6.64 \approx 7$$

The pole positions are:

$$s_k = \Omega_c e^{j \left[\frac{\pi}{2} + (2k+1)\pi/2N \right]}$$

$$= 1000 \pi e^{j \left[\frac{\pi}{2} + (2k+1)\pi/14 \right]}, \quad k = 0, 1, 2, 3, 4, 5, 6$$

where Ω_c is 3 dB cutoff frequency.

EXAMPLE 8.25 Determine the order of a Butterworth low-pass filter satisfying the following specifications:

$$f_p = 0.10 \text{ Hz}, \quad \alpha_p = 0.5 \text{ dB}$$

$$f_s = 0.15 \text{ Hz}, \quad \alpha_s = 15 \text{ dB}; f = 1 \text{ Hz}$$

Solution: Given

$$f_p = 0.10 \text{ Hz}, \quad \therefore \omega_p = \omega_1 = 2\pi f_p = 2\pi(0.1) = 0.2\pi$$

$$f_s = 0.15 \text{ Hz}, \quad \therefore \omega_s = \omega_2 = 2\pi f_s = 2\pi(0.15) = 0.30\pi$$

$$\alpha_p = \alpha_1 = 0.5 \text{ dB}, \quad \therefore k_1 = -0.5 \text{ dB}, \text{ so } A_1 = 10^{k_1/20} = 10^{-0.5/20} = 0.944$$

$$\alpha_s = \alpha_2 = 15 \text{ dB}, \quad \therefore k_2 = -15 \text{ dB}, \text{ so } A_2 = 10^{k_2/20} = 10^{-15/20} = 0.177$$

$$f = 1 \text{ Hz}, \quad \therefore T = \frac{1}{f} = \frac{1}{1} = 1 \text{ s.}$$

1. The type of transformation is not specified. Let us use bilinear transformation.

$$2. \quad \frac{\Omega_2}{\Omega_1} = \frac{\frac{2}{T} \tan \frac{\omega_2}{2}}{\frac{2}{T} \tan \frac{\omega_1}{2}} = \frac{\frac{1}{1} \tan \frac{0.3\pi}{2}}{\frac{1}{1} \tan \frac{0.2\pi}{2}} = \frac{1.019}{0.649} = 1.57$$

$$3. \quad N \geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{A_2^2} - 1 \right] / \left[\frac{1}{A_1^2} - 1 \right] \right\}}{\log \left(\frac{\Omega_2}{\Omega_1} \right)}$$

$$\geq \frac{1}{2} \frac{\log \left\{ \left[\frac{1}{0.177^2} - 1 \right] / \left[\frac{1}{0.944^2} - 1 \right] \right\}}{\log (1.57)}$$

$$\geq 6.16 \approx 7$$

so the order of the low-pass Butterworth filter is $N = 7$.